Student Submission: Divide-and-Conquer Analysis

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Derivation Steps:

1. The standard Master Theorem representation is $T(n) = aT(n/b) + f(n)$.
2. Comparing with $T(n) = 2T(n/2) + \Theta(n)$, we obtain parameters $a = 2, b = 2, f(n) = \Theta(n)$.
3. We compute the critical exponent:

$$\log_b a = \log_2 2 = 1 \implies n^{\log_b a} = n^1 = n$$

4. Since $f(n) = \Theta(n) = \Theta(n^{\log_b a})$, this recurrence strictly satisfies **Case 2** of the Master Theorem.
5. Therefore, the asymptotic tight bound is:

$$T(n) = \Theta(n^{\log_b a} \log n) = \Theta(n \log n)$$